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# MieNet: A Python package for fast well-mixed cloud particle opacity calculations

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**Software**

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## Summary

`MieNet` is a Python package for quick calculations of well-mixed cloud particle opacities. `MieNet` produces the extinction, scattering, and asymmetry Mie coefficients given the wavelength, cloud particle radius, and cloud species with their respective volume fraction (VMR). Cloud particles are assumed to be well-mixed and spherical. Artificial neural networks (ANNs), made with `TensorFlow` (Abadi et al., 2016), are used to overcome the computational challenge associated with well-mixed cloud particles. There are three calculation methods available: full calculations, grid interpolations, or ANN predictions.

## Statement of Need

Modeling exoplanet clouds is necessary to better understand the formation of exoplanets and to interpret their observations (Christiane Helling, 2019; Powell et al., 2019). Cloud models require assumptions to be made about the shape (M I Mishchenko & Travis (editors), 2000; M. Min et al., 2003, 2006; Samra et al., 2020), formation (Gao & Benneke, 2018; Christiane Helling, 2019) and composition (Gao & Powell (2021); Sven Kiefer et al. (2024); Kiefer et al. subm.) of cloud particles. Exoplanet clouds are often assumed to be homogeneous, or comprised of only one material, but theoretical models (Ch Helling & Woitke, 2006; Lee, 2023; Michiel Min et al., 2020; Ormel & Min, 2019) have predicted and some observations (Dyrek et al., 2024; Hoch et al., 2025), show evidence of heterogeneous cloud particles. However, modeling these heterogeneous clouds is computationally intensive.

Mie theory (Gustav, 1908) uses an effective refractive index and size parameter, which is related to the cloud particle radius and wavelength of light. Mie scattering calculations are already a slow process, but for heterogeneous cloud particles, calculating the effective refractive indices adds additional computation time. The high dimensionality (wavelength, radius, VMR of Species 1, VMR of Species 2, etc.) of mixed Mie particles results in a large parameter space which makes grid calculations unfeasible when high resolutions are required. These computational challenges caused by calculating the effective refractive index make modeling the mixed particles observed in brown dwarfs (Dyrek et al., 2024; Hoch et al., 2025) and strongly mixed clouds predicted by theoretical models (Christiane Helling, 2019; Christiane Helling et al., 2023; Samra et al., 2020) extremely intensive. Retrievals require a large number of models, so each run needs to be fast. Even a small increase in speed for each run vastly improves computation time, which is why machine learning techniques are necessary.

## State of the Field

Other studies have successfully applied neural networks to Mie theory and are able to make fast and accurate opacity calculations. However, their work utilizes different assumptions and is not

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applicable to well-mixed cloud particles. `MieAi` (Kumar et al., 2024) and `NeuralMie` (Geiss & Ma, 2025) both assume core-shell configuration and are intended for Earth climate and weather modeling. `Glitterin` (Lin et al., 2025) focuses on debris disks and assumes non-spherical particles. `MieNet` is unique in its assumption of well-mixed, spherical cloud particles. It utilizes neural networks to overcome the computational challenge of effective refractive indices and optimizes well-mixed opacity calculations.

## Software Design

`MieNet` is designed with a focus on computational speed, easy integration, and adaptability. Functions to produce Mie coefficients are:

- `efficiencies`: fully calculate coefficients with user-defined mixing theory and Mie theory
- `ai_efficiencies`: use ANNs to predict coefficients
- `grid_efficiencies`: interpolate coefficients from a pre-calculated grid
- `auto_efficiencies`: choose the fastest available method (full, AI, or grid) to produce coefficients

All functions are easily interchangeable since they have the same input and output structure. The outputs are also similar to other Mie-theory codes, such as `miepython` (Prahl, 2026) or `PyMieScatt` (Sumlin et al., 2018), so working with `MieNet` is familiar and allows for easy integration of `MieNet` into existing code.

For the `efficiencies` function, two mixing theories are available: Landau, Lifshitz, and Looyenga (LLL) (Landau, 1960; Looyenga, 1965) and Bruggeman (Bruggeman, 1935). Mixing theories evaluate the effective refractive index of a well-mixed particle. LLL and Bruggeman are the most common used in the field and have both been shown to match lab results in different setups (Kolokolova & Gustafson, 2001; Thomas & Gautier, 2009; Voshchinnikov et al., 2007). While these theories are similar, there are some differences (see Sven Kiefer et al., 2024). `MieNet`'s current ANNs are trained on datasets made using the LLL approximation, because it is less prone to computational artifacts.

Users can create their own training set with either mixing theory using the `generate_training_set` function. Once a training set is made, one can use `train_ai_model` to train and save custom ANNs. These functionalities allows users to easily train `MieNet`-compatible ANNs for specific mixtures, wavelength and particle size ranges, or mixing theory that `MieNet`'s current ANNs do not cover. For a further discussion on how `MieNet`'s ANNs are trained, see Kiefer et al. subm.

To create grids for the `grid_efficiencies` function, `MieNet` contains `produce_efficiency_grid` and `load_grid_efficiency` to allow users to precalculate and load grids. However, these grids are limited to lower resolution, as the dataset size required for a high-accuracy grid is too large.

## Research Impact Statement

Mie theory is used for calculating the opacity of exoplanet clouds by the vast majority of the field (Grant et al., 2023; S. Kiefer et al., 2024; Powell et al., 2019; Steinrueck et al., 2025). `MieNet`'s use of ANNs and optimization of mixed particle Mie calculations makes the inclusion and study of heterogeneous cloud particles feasible for the research community. This package has already shown direct impact in works that have investigated interplanetary dust particle (IDP) infall as an exoplanet cloud seeding mechanism (Sven Kiefer, Gosselin, et al., 2026) and the mixing state of YSES-1c's cloud particles (Kiefer et al. subm.). `MieNet` is integrated into the `Virga` (Batalha et al. (2026); Moran et al. (2025); Kiefer et al. subm.) and `PICASO` (Batalha et al., 2019; Mang et al., 2026; Mukherjee et al., 2023) codes, which are publicly available, open-source atmospheric models commonly used by the exoplanet community (Ahrer et al., 2023; Bardalez Gagliuffi et al., 2025; Beichman et al., 2025; Hamill et al., 2024; He et

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al., 2024; Sven Kiefer, Morley, et al., 2026; Mukherjee et al., 2026). The similarity of MieNet's outputs to other common Mie-theory libraries, like `miepython` (Prahl, 2026) or `PyMieScatt` (Sumlin et al., 2018), allows for easy integration of heterogeneous particles into other cloud and planetary codes which use these libraries (Bhattacharjee, 2025; Espinoza et al., 2024; Pater et al., 2026; Shefferson-Nagata & Ohno, 2026; Xie et al., 2025). While this package was developed for exoplanet atmospheres, `MieNet` can be utilized in other areas that also use Mie theory, such as climate and weather (Chakrabarty et al., 2023,; Fierce et al., 2020; Ji et al., 2026; Karlsson et al., 2021), protoplanetary and debris disks (Andrews et al., 2011; Kranhold et al., 2026; Wang et al., 2021; Xie et al., 2025), solar-system (Dai et al., 2024; Hedman & MacKenzie, 2026; Pater et al., 2026; Vidwans et al., 2022), interstellar medium dust (Compiègne et al., 2011; De Looze et al., 2017; Galliano et al., 2011; Guerlet et al., 2014), optical sensing and imaging (Carrico et al., 2021; Hagan & Kroll, 2020; Thiele et al., 2024; Tryner et al., 2020) and other applications.

## AI Usage Disclosure

AI tools were used during software development for debugging. No generative AI tools were used in the writing of this manuscript or MieNet documentation.

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## References


Abadi, M., Agarwal, A., Barham, P., Brevdo, E., Chen, Z., Citro, C., Corrado, G. S., Davis, A., Dean, J., Devin, M., Ghemawat, S., Goodfellow, I., Harp, A., Irving, G., Isard, M., Jia, Y., Jozefowicz, R., Kaiser, L., Kudlur, M., … Zheng, X. (2016). *TensorFlow: Large-Scale Machine Learning on Heterogeneous Distributed Systems*.

Ahrer, E.-M., Stevenson, K. B., Mansfield, M., Moran, S. E., Brande, J., Morello, G., Murray, C. A., Nikolov, N. K., Petit dit de la Roche, D. J. M., Schlawin, E., Wheatley, P. J., Zieba, S., Batalha, N. E., Damiano, M., Goyal, J. M., Lendl, M., Lothringer, J. D., Mukherjee, S., Ohno, K., … Zhang, X. (2023). Early Release Science of the exoplanet WASP-39b with JWST NIRCam. *Nature*, *614*(7949), 653–658. https://doi.org/10.1038/s41586-022-05590-4

Andrews, S., Wilner, D., Espaillat, C., Hughes, A. M., Dullemond, C. P., McClure, M. K., Qi, C., & Brown, J. M. (2011). Resolved Images of Large Cavities in Protoplanetary Transition Disks. *The Astrophysical Journal*, *732*(1), 42. https://doi.org/10.1088/0004-637X/732/1/42

Bardalez Gagliuffi, D. C., Balmer, W. O., Pueyo, L., Brandt, T. D., Giovinazzi, M. R., Millholland, S., Black, B., Lu, T., Rice, M., Mang, J., Morley, C., Lacy, B., Girard, J. H., Matthews, E. C., Carter, A. L., Bowler, B. P., Faherty, J. K., Fontanive, C., & Rickman, E. (2025). JWST Coronagraphic Images of 14 Her c: A Cold Giant Planet in a Dynamically Hot Multiplanet System. *The Astrophysical Journal Letters*, *988*(1), L18. https://doi.org/10.3847/2041-8213/ade30f

Batalha, N. E., Marley, M. S., Lewis, N. K., & Fortney, J. J. (2019). Exoplanet Reflected-light Spectroscopy with PICASO. *The Astrophysical Journal*, *878*(1), 70. https://doi.org/10.3847/1538-4357/ab1b51

Batalha, N. E., Rooney, C. M., Visscher, C., Moran, S. E., Marley, M. S., Sengupta, A. R., Kiefer, S., Lodge, M. G., Mang, J., Morley, C. V., Mukherjee, S., Fortney, J. J., Gao, P., Lewis, N. K., Mayorga, L. C., Pearce, L. A., & Wakeford, H. R. (2026). Condensation Clouds in Substellar Atmospheres with Virga. *The Astronomical Journal*, *171*(2), 98. https://doi.org/10.3847/1538-3881/ae29e5


DRAFT

Beichman, C., Sanghi, A., Mawet, D., Kervella, P., Wagner, K., Quarles, B., Lissauer, J. J., Sommer, M., Wyatt, M., Godoy, N., Balmer, W. O., Pueyo, L., Llop-Sayson, J., Aguilar, J., Akeson, R., Belikov, R., Boccaletti, A., Choquet, E., Fomalont, E., … Zilinskas, M. (2025). Worlds Next Door: A Candidate Giant Planet Imaged in the Habitable Zone of α Centauri A. I. Observations, Orbital and Physical Properties, and Exozodi Upper Limits. *The Astrophysical Journal Letters*, *989*(2), L22. https://doi.org/10.3847/2041-8213/adf53f

Bhattacharjee, S. (2025). Thick Disks Around White Dwarfs Viewed "Edge-off": Effects on Transit Properties and Infrared Excess. *Publications of the Astronomical Society of the Pacific*, *137*(11), 114204. https://doi.org/10.1088/1538-3873/ae0fe3

Bruggeman, D. a. G. (1935). Berechnung verschiedener physikalischer Konstanten von heterogenen Substanzen. I. Dielektrizitätskonstanten und Leitfähigkeiten der Mischkörper aus isotropen Substanzen. *Annalen Der Physik*, *416*(7), 636–664. https://doi.org/https://doi.org/10.1002/andp.19354160705

Carrico, C. M., Capek, T. J., Gorkowski, K. J., Lam, J. T., Gulick, S., Karacaoglu, J., Lee, J. E., Dungan, C., Aiken, A. C., Onasch, T. B., Freedman, A., Mazzoleni, C., & Dubey, M. K. (2021). Humidified single-scattering albedometer (H-CAPS-PMSSA): Design, data analysis, and validation. *Aerosol Science Technology*, *55*, 749–768. https://doi.org/10.1080/02786826.2021.1895430

Chakrabarty, R. K., Shetty, N. J., Thind, A. S., Beeler, P., Sumlin, B. J., Zhang, C., Liu, P., Idrobo, J. C., Adachi, K., Wagner, N. L., Schwarz, J. P., Ahern, A., Sedlacek, A. J., Lambe, A., Daube, C., Lyu, M., Liu, C., Herndon, S., Onasch, T. B., & Mishra, R. (2023). Shortwave absorption by wildfire smoke dominated by dark brown carbon. *Nature Geoscience*, *16*, 683–688. https://doi.org/10.1038/s41561-023-01237-9

Compiègne, M., Verstraete, L., Jones, A., Bernard, J.-P., Boulanger, F., Flagey, N., Le Bourlot, J., Paradis, D., & Ysard, N. (2011). The global dust SED: Tracing the nature and evolution of dust with DustEM. *Astronomy and Astrophysics*, *525*, A103. https://doi.org/10.1051/0004-6361/201015292

Dai, L., Shao, W., & Sheng, Z. (2024). An investigation into Venusian atmospheric chemistry based on an open-access photochemistry-transport model at 0 112 km. *Astronomy and Astrophysics*, *689*, A55. https://doi.org/10.1051/0004-6361/202450552

De Looze, I., Barlow, M. J., Swinyard, B. M., Rho, J., Gomez, H. L., Matsuura, M., & Wesson, R. (2017). The dust mass in Cassiopeia A from a spatially resolved Herschel analysis. *Monthly Notices of the Royal Astronomical Society*, *465*, 3309–3342. https://doi.org/10.1093/mnras/stw2837

Dyrek, A., Min, M., Decin, L., Bouwman, J., Crouzet, N., Mollière, P., Lagage, P.-O., Konings, T., Tremblin, P., Güdel, M., Pye, J., Waters, R., Henning, T., Vandenbussche, B., Ardevol Martinez, F., Argyriou, I., Ducrot, E., Heinke, L., Looveren, G. van, … Wright, G. (2024). SO2, silicate clouds, but no CH4 detected in a warm Neptune. *Nature*, *625*(7993), 51–54. https://doi.org/10.1038/s41586-023-06849-0

Espinoza, N., Steinrueck, M. E., Kirk, J., MacDonald, R. J., Savel, A. B., Triantafillides, A., Kempton, E. M.-R., Murphy, M. M., Carone, L., Zamyatina, M., Lewis, D. A., Samra, D., Kiefer, S., Rauscher, E., Christie, D., Mayne, N., Helling, C., Rustamkulov, Z., Parmentier, V., … Crouzet, N. (2024). Inhomogeneous terminators on the exoplanet WASP-39 b. *Nature*, *632*, 1017–1020. https://doi.org/10.1038/s41586-024-07768-4

Fierce, L., Onasch, T. B., Cappa, C. D., Mazzoleni, C., China, S., Bhandari, J., Davidovits, P., Al Fischer, D., Helgestad, T., Lambe, A. T., Sedlacek, A. J., Smith, G. D., & Wolff, L. (2020). Radiative absorption enhancements by black carbon controlled by particle-to-particle heterogeneity in composition. *Proceedings of the National Academy of Science*, *117*, 5196–5203. https://doi.org/10.1073/pnas.1919723117

DRAFT

Galliano, F., Hony, S., Bernard, J.-P., Bot, C., Madden, S. C., Roman-Duval, J., Galametz, M., Li, A., Meixner, M., Engelbracht, C. W., Lebouteiller, V., Misselt, K., Montiel, E., Panuzzo, P., Reach, W. T., & Skibba, R. (2011). Non-standard grain properties, dark gas reservoir, and extended submillimeter excess, probed by Herschel in the Large Magellanic Cloud. *Astronomy and Astrophysics*, *536*, A88. https://doi.org/10.1051/0004-6361/201117952

Gao, P., & Benneke, B. (2018). Microphysics of KCl and ZnS Clouds on GJ 1214 b. *The Astrophysical Journal*, *863*(2), 165. https://doi.org/10.3847/1538-4357/aad461

Gao, P., & Powell, D. (2021). A Universal Cloud Composition on the Nightsides of Hot Jupiters. *The Astrophysical Journal*, *918*, L7. https://doi.org/10.3847/2041-8213/ac139f

Geiss, A., & Ma, P.-L. (2025). NeuralMie (v1.0): An aerosol optics emulator. *Geoscientific Model Development*, *18*(5), 1809–1827. https://doi.org/10.5194/gmd-18-1809-2025

Grant, D., Lewis, N. K., Wakeford, H. R., Batalha, N. E., Glidden, A., Goyal, J., Mullens, E., MacDonald, R. J., May, E. M., Seager, S., Stevenson, K. B., Valenti, J. A., Visscher, C., Alderson, L., Allen, N. H., Cañas, C. I., Colón, K., Clampin, M., Espinoza, N., … Watkins, L. L. (2023). JWST-TST DREAMS: Quartz Clouds in the Atmosphere of WASP-17b. *The Astrophysical Journal*, *956*, L32. https://doi.org/10.3847/2041-8213/acfc3b10.3847/2041-8213/acfdab

Guerlet, S., Spiga, A., Sylvestre, M., Indurain, M., Fouchet, T., Leconte, J., Millour, E., Wordsworth, R., Capderou, M., Bézard, B., & Forget, F. (2014). Global climate modeling of Saturn's atmosphere. Part I: Evaluation of the radiative transfer model. *Icarus*, *238*, 110–124. https://doi.org/10.1016/j.icarus.2014.05.010

Gustav, M. (1908). *Beitrage zur Optik truber Medien, speziell kolloidaler Metallosungen [Optics of cloudy media, especially colloidal metal solutions]*. https://hero.epa.gov/reference/39703/

Hagan, D. H., & Kroll, J. H. (2020). Assessing the accuracy of low-cost optical particle sensors using a physics-based approach. *Atmospheric Measurement Techniques*, *13*, 6343–6355. https://doi.org/10.5194/amt-13-6343-2020

Hamill, C. D., Johnson, A. V., Batalha, N., Nag, R., Gao, P., Adams, D., & Kataria, T. (2024). Reflected-light Phase Curves with PICASO: A Kepler-7b Case Study. *The Astrophysical Journal*, *976*(2), 181. https://doi.org/10.3847/1538-4357/ad7de6

He, C., Radke, M., Moran, S. E., Hörst, S. M., Lewis, N. K., Moses, J. I., Marley, M. S., Batalha, N. E., Kempton, E. M.-R., Morley, C. V., Valenti, J. A., & Vuitton, V. (2024). Optical properties of organic haze analogues in water-rich exoplanet atmospheres observable with JWST. *Nature Astronomy*, *8*, 182–192. https://doi.org/10.1038/s41550-023-02140-4

Hedman, M. M., & MacKenzie, S. M. (2026). *Features in the visible spectra of the Enceladus particle plume and E ring: Potential evidence of organic materials and/or missing sub-micron particles*. arXiv. https://doi.org/10.48550/arXiv.2607.15940

Helling, Christiane. (2019). Exoplanet Clouds. *Annual Review of Earth and Planetary Sciences*, *47*(1), 583–606. https://doi.org/10.1146/annurev-earth-053018-060401

Helling, Christiane, Samra, D., Lewis, D., Calder, R., Hirst, G., Woitke, P., Baeyens, R., Carone, L., Herbort, O., & Chubb, K. L. (2023). Exoplanet weather and climate regimes with clouds and thermal ionospheres - A model grid study in support of large-scale observational campaigns. *Astronomy & Astrophysics*, *671*, A122. https://doi.org/10.1051/0004-6361/202243956

Helling, Ch, & Woitke, P. (2006). Dust in brown dwarfs - V. Growth and evaporation of dirty dust grains. *Astronomy & Astrophysics*, *455*(1), 325–338. https://doi.org/10.1051/0004-6361:20054598

Hoch, K. K. W., Rowland, M., Petrus, S., Nasedkin, E., Ingebretsen, C., Kammerer, J., Perrin,

DRAFT

M., D'Orazi, V., Balmer, W. O., Barman, T., Bonnefoy, M., Chauvin, G., Chen, C., De Rosa, R. J., Girard, J., Gonzales, E., Kenworthy, M., Konopacky, Q. M., Macintosh, B., … Zhang, Y. (2025). Silicate clouds and a circumplanetary disk in the YSES-1 exoplanet system. *Nature*, *643*(8073), 938–942. https://doi.org/10.1038/s41586-025-09174-w

Ji, Y., Kong, L., Wang, J., Zhao, B., Zhang, Z., Chen, S., Lin, S., Chen, J., Gao, Y., & Ji, Y. (2026). Mechanism of heterogeneous mixing between typical oxygenated volatile organic compound and hematite: Implication for radiative forcing. *Atmospheric Environment*, *380*, 122089. https://doi.org/10.1016/j.atmosenv.2026.122089

Karlsson, L., Krejci, R., Koike, M., Ebell, K., & Zieger, P. (2021). A long-term study of cloud residuals from low-level Arctic clouds. *Atmospheric Chemistry & Physics*, *21*, 8933–8959. https://doi.org/10.5194/acp-21-8933-2021

Kiefer, S., Bach-Møller, N., Samra, D., Lewis, D. A., Schneider, A. D., Amadio, F., Lecoq-Molinos, H., Carone, L., Decin, L., Jørgensen, U. G., & Helling, C. (2024). Under the magnifying glass: A combined 3D model applied to cloudy warm Saturn-type exoplanets around M dwarfs. *Astronomy & Astrophysics*, *692*, A222. https://doi.org/10.1051/0004-6361/202450753

Kiefer, Sven, Gosselin, G. J., Morley, C. V., & Gulick, S. P. S. (2026). *Interplanetary Dust Ablation Seeds Silicate Clouds in Exoplanet Atmospheres*. arXiv. https://doi.org/10.48550/arXiv.2609.08960

Kiefer, Sven, Morley, C. V., & Rowland, M. J. (2026). Connecting JWST Silicate Cloud Observations to Exoplanet Cloud Microphysics with Nimbus. *The Astrophysical Journal*, *1001*(1), 98. https://doi.org/10.3847/1538-4357/ae5101

Kiefer, Sven, Samra, D., Lewis, D. A., Schneider, A. D., Min, M., Carone, L., Decin, L., & Helling, C. (2024). Why heterogeneous cloud particles matter: Iron-bearing species and cloud particle morphology affects exoplanet transmission spectra. *Astronomy & Astrophysics*, *690*, A244. https://doi.org/10.1051/0004-6361/202450526

Kolokolova, L., & Gustafson, B. A. S. (2001). Scattering by inhomogeneous particles: Microwave analog experiments and comparison to effective medium theories. *JQSRT*, *70*(4), 611–625. https://doi.org/10.1016/S0022-4073(01)00033-4

Kranhold, C., Sadeghi, A., Harries, D., Mutschke, H., Vegesna, S. V., Langenhorst, F., Stanca, S. E., Plentz, J., & Krüger, H. (2026). Iron(II) sulphides in cosmic dust: Optical constants depending on iron deficiency and temperature. *Astronomy and Astrophysics*, *711*, A106. https://doi.org/10.1051/0004-6361/202659441

Kumar, P., Vogel, H., Bruckert, J., Muth, L. J., & Hoshyaripour, G. A. (2024). MieAI: A neural network for calculating optical properties of internally mixed aerosol in atmospheric models. *Npj Climate and Atmospheric Science*, *7*(1), 110. https://doi.org/10.1038/s41612-024-00652-y

Landau, (Lev. D. (1960). *Electrodynamics of continuous media*. Oxford, New York, Pergamon Press. http://archive.org/details/electrodynamicso00land

Lee, E. K. H. (2023). Modelling dynamically driven global cloud formation microphysics in the HAT-P-1b atmosphere. *Monthly Notices of the Royal Astronomical Society*, *524*(2), 2918–2933. https://doi.org/10.1093/mnras/stad2037

Lin, Z.-Y. D., Weinberger, A. J., Zubko, E., Arnold, J. A., & Videen, G. (2025). *Glitterin: Towards Replacing the Role of Lorenz-Mie Theory in Astronomy Using Neural Networks Trained on Light Scattering of Irregularly Shaped Grains*. arXiv. https://doi.org/10.48550/arXiv.2511.09668

Looyenga, H. (1965). Dielectric constants of heterogeneous mixtures. *Physica*, *31*(3), 401–406. https://doi.org/10.1016/0031-8914(65)90045-5

DRAFT

M I Mishchenko, J. W. H., & Travis (editors), L. D. (2000). Light Scattering by Nonspherical Particles: Theory, Measurements, and Applications. *Measurement Science and Technology*, *11*(12), 1827. https://doi.org/10.1088/0957-0233/11/12/705

Mang, J., Batalha, N. E., Morley, C. V., Wogan, N. F., Mukherjee, S., Visscher, C., Marley, M. S., Fortney, J. J., Chubb, K. L., Gao, P., & Malsky, I. (2026). PICASO 4.0: Clouds and Photochemistry in Climate Models of Brown Dwarfs and Exoplanets. *The Astrophysical Journal*, *1000*(1), 98. https://doi.org/10.3847/1538-4357/ae47ff

Min, M., Hovenier, J. W., Dominik, C., Koter, A. de, & Yurkin, M. A. (2006). Absorption and scattering properties of arbitrarily shaped particles in the Rayleigh domain. *Journal of Quantitative Spectroscopy and Radiative Transfer*, *97*, 161–180. https://doi.org/10.1016/j.jqsrt.2005.05.059

Min, M., Hovenier, J. W., & Koter, A. de. (2003). Shape effects in scattering and absorption by randomly oriented particles small compared to the wavelength. *Astronomy & Astrophysics*, *404*(1), 35–46. https://doi.org/10.1051/0004-6361:20030456

Min, Michiel, Ormel, C. W., Chubb, K., Helling, C., & Kawashima, Y. (2020). The ARCiS framework for exoplanet atmospheres - Modelling philosophy and retrieval. *Astronomy & Astrophysics*, *642*, A28. https://doi.org/10.1051/0004-6361/201937377

Moran, S. E., Lodge, M. G., Batalha, N. E., Ohno, K., Vahidinia, S., Marley, M. S., Wakeford, H. R., & Leinhardt, Z. M. (2025). Fractal Aggregate Aerosols in the Virga Cloud Code. I. Model Description and Application to a Benchmark Cloudy Exoplanet. *The Astrophysical Journal*, *994*(1), 116. https://doi.org/10.3847/1538-4357/ae0583

Mukherjee, S., Batalha, N. E., Fortney, J. J., & Marley, M. S. (2023). PICASO 3.0: A One-dimensional Climate Model for Giant Planets and Brown Dwarfs. *The Astrophysical Journal*, *942*, 71. https://doi.org/10.3847/1538-4357/ac9f48

Mukherjee, S., Nixon, M. C., Welbanks, L., Mang, J., Wogan, N. F., Batalha, N. E., & Line, M. R. (2026). Impact of Clouds on the Atmosphere–Mantle Interface of Sub-Neptunes. *The Astrophysical Journal Letters*, *1005*(2), L63. https://doi.org/10.3847/2041-8213/ae7432

Ormel, C. W., & Min, M. (2019). ARCiS framework for exoplanet atmospheres - The cloud transport model. *Astronomy & Astrophysics*, *622*, A121. https://doi.org/10.1051/0004-6361/201833678

Pater, I. de, Showalter, M., Hedman, Matt, & Hammel, H. (2026). *Characterization of the Outer Uranian Rings in the Visual and Near-IR Using Keck, JWST, and HST Observations - Pater - 2026 - Journal of Geophysical Research: Planets - Wiley Online Library*. https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JE009404

Powell, D., Louden, T., Kreidberg, L., Zhang, X., Gao, P., & Parmentier, V. (2019). Transit Signatures of Inhomogeneous Clouds on Hot Jupiters: Insights from Microphysical Cloud Modeling. *The Astrophysical Journal*, *887*, 170. https://doi.org/10.3847/1538-4357/ab55d9

Prahl, S. (2026). *Miepython: Pure python calculation of Mie scattering*. Zenodo. https://doi.org/10.5281/zenodo.18893972

Samra, D., Helling, C., & Min, M. (2020). Mineral snowflakes on exoplanets and brown dwarfs: Effects of micro-porosity, size distributions, and particle shape. *Astronomy & Astrophysics*, *639*, A107. https://doi.org/10.1051/0004-6361/202037553

Shefferson-Nagata, T. M., & Ohno, K. (2026). Radiatively Controlled Thermal Stability of High-altitude Clouds in Exoplanetary Atmospheres. *The Astrophysical Journal*, *1005*(2), 210. https://doi.org/10.3847/1538-4357/ae7d27

Steinrueck, M. E., Savel, A. B., Christie, D. A., Carone, L., Tsai, S.-M., Akın, C., Kennedy, T. D., Kiefer, S., Lewis, D. A., Rauscher, E., Samra, D., Zamyatina, M., Arnold, K., Baeyens,

DRAFT

R., Gkouvelis, L., Haegele, D., Helling, C., Mayne, N. J., Powell, D., … Tan, X. (2025). *Limb Asymmetries on WASP-39b: A Multi-GCM Comparison of Chemistry, Clouds, and Hazes*. arXiv. https://doi.org/10.48550/arXiv.2509.21588

Sumlin, B. J., Heinson, W. R., & Chakrabarty, R. K. (2018). Retrieving the aerosol complex refractive index using PyMieScatt: A Mie computational package with visualization capabilities. *Journal of Quantitative Spectroscopy and Radiative Transfer*, *205*, 127–134. https://doi.org/10.1016/j.jqsrt.2017.10.012

Thiele, J. C., Pfitzner, E., & Kukura, P. (2024). Single-protein optical holography. *Nature Photonics*, *18*, 388–395. https://doi.org/10.1038/s41566-024-01405-2

Thomas, M., & Gautier, C. (2009). Investigations of the March 2006 African dust storm using ground-based column-integrated high spectral resolution infrared (8–13 µm) and visible aerosol optical thickness measurements: 2. Mineral aerosol mixture analyses. *J. Geophys. Res. Atmos.*, *114*(D14). https://doi.org/10.1029/2008JD010931

Tryner, J., Mehaffy, J., Miller-Lionberg, D., & Volckens, J. (2020). Effects of aerosol type and simulated aging on performance of low-cost PM sensors. *Journal of Aerosol Science*, *150*, 105654. https://doi.org/10.1016/j.jaerosci.2020.105654

Vidwans, A., Choudhary, S., Jolliff, B., Gillis-Davis, J., & Biswas, P. (2022). Size and charge distribution characteristics of fine and ultrafine particles in simulated lunar dust: Relevance to lunar missions and exploration. *Planetary and Space Science*, *210*, 105392. https://doi.org/10.1016/j.pss.2021.105392

Voshchinnikov, N. V., Videen, G., & Henning, T. (2007). Effective medium theories for irregular fluffy structures: Aggregation of small particles. *Appl. Opt., AO*, *46*(19), 4065–4072. https://doi.org/10.1364/AO.46.004065

Wang, J. J., Vigan, A., Lacour, S., Nowak, M., Stolker, T., De Rosa, R. J., Ginzburg, S., Gao, P., Abuter, R., Amorim, A., Asensio-Torres, R., Bauböck, M., Benisty, M., Berger, J. P., Beust, H., Beuzit, J.-L., Blunt, S., Boccaletti, A., Bohn, A., … The GRAVITY Collaboration. (2021). Constraining the Nature of the PDS 70 Protoplanets with VLTI/GRAVITY*. *The Astronomical Journal*, *161*(3), 148. https://doi.org/10.3847/1538-3881/abdb2d

Xie, C., Chen, C. H., Lisse, C. M., Hines, D. C., Beck, T., Betti, S. K., Pinilla-Alonso, N., Ingebretsen, C., Worthen, K., Gáspár, A., Wolff, S. G., Bolin, B. T., Pueyo, L., Perrin, M. D., Stansberry, J. A., & Leisenring, J. M. (2025). Water ice in the debris disk around HD 181327. *Nature*, *641*(8063), 608–611. https://doi.org/10.1038/s41586-025-08920-4

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